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4 la
mps containing only gases of class 2 .2 (a
ccording to 2 .2 .2 .2) p
rovided they are packaged so that the
projectile effects of any rupture of the bulb will be contained within the package .
1.1.1.10 De vices containing dangerous goods, which are in use or intended for use during transport
For devices containing dangerous goods, which are in use or intended for use during transport, see 5 .5 .4 .
1.1.2 C onventions
1.1.2.1 In ternational Convention for the Safety of Life at Sea, 1974
Part A of chapter VII of the International Convention for the Safety of Life at Sea, 1974 (SOLAS), as amended, deals with the carriage of dangerous goods in packaged form, and is reproduced in full:
RESOLUTION MSC.556(108) (adopted on 23 May 2024)
AMENDMENTS TO THE
INTERNATIONAL MARITIME DANGEROUS GOODS (IMDG) CODE

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RESOLUTION MSC.565(108) (adopted on 24 May 2024)
REVISED INTERIM RECOMMENDATIONS FOR CARRIAGE OF
LIQUEFIED HYDROGEN IN BULK
MSC 108/20/Add.1
Annex 20, page 1
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ANNEX 20
RESOLUTION MSC.565(108)
(adopted on 24 May 2024)
REVISED INTERIM RECOMMENDATIONS FOR CARRIAGE OF
LIQUEFIED HYDROGEN IN BULK
THE MARITIME SAFETY COMMITTEE,
RECALLING Article 28(b) of the Convention on the International Maritime Organization concerning the functions of the Committee,
NOTING that the International Convention for the Safety of Life at Sea ("the Convention"), 1974 and the International Code for the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk ("the IGC Code") currently do not specifically provide requirements for carriage of liquefied hydrogen in bulk by sea,
NOTING ALSO that paragraph 5 of Preamble of the IGC Code states that requirements for new products and their conditions of carriage will be circulated as recommendations, on an
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new products and their conditions of carriage will be circulated as recommendations, on an interim basis, prior to the entry into force of the appropriate amendments,
RECOGNIZING a need for the development of the Revised interim recommendations for carriage of liquefied hydrogen in bulk,
ACKNOWLEDGING that, in the interim, there is an urgent need to provide recommendations to the Administrations on safe carriage of liquefied hydrogen in bulk,
ACKNOWLEDGING ALSO that the Revised Interim Recommendations are intended to facilitate establishment of a tripartite agreement for a pilot ship, which will be developed for the

research and demonstration of safe long-distance overseas carriage of liquefied hydrogen in bulk,
HAVING CONSIDERED the Revised Interim Recommendations prepared by the Sub - Committee on Carriage of Cargoes and Containers at its ninth session,
1 ADOPTS the Revised interim recommendations for carriage of liquefied hydrogen in
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1 ADOPTS the Revised interim recommendations for carriage of liquefied hydrogen in bulk, the text of which is set out in the annex to the present resolution;
2 INVITES Member States to apply the Revised Interim Recommendations to the pilot ship carrying liquefied hydrogen in bulk taking the explanatory notes into consideration;
3 AGREES to acquire information on safe carriage of liquefied hydrogen in bulk prior to amendment to the IGC Code for the inclusion of liquefied hydrogen;
4 ALSO AGREES that these Revised Interim Recommendations may need to be reviewed if they are to be applied to ships other than the pilot ship; and
5 URGES Member States and the industry to submit information, observations, comments and recommendations based on the practical experience gained through the application of the Revised Interim Recommendations and submit relevant safety analysis on ships carrying liquefied hydrogen in bulk.
6 REVOKES resolution MSC.420(97).

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on ships carrying liquefied hydrogen in bulk.
6 REVOKES resolution MSC.420(97).
RESOLUTION MSC.565(108) (adopted on 24 May 2024)
REVISED INTERIM RECOMMENDATIONS FOR CARRIAGE OF LIQUEFIED HYDROGEN IN BULK

ANNEX
REVISED INTERIM RECOMMENDATIONS FOR CARRIAGE OF LIQUEFIED HYDROGEN IN BULK
1 INTRODUCTION
1.1 For the carriage of liquefied gases in bulk by ships, the ships should comply with the relevant requirements in the IGC Code, as amended ("the Code"). The scope of the Code provided in paragraph 1.1.1 is:
"The Code applies to ships regardless of their size, including those of less than 500 gross tonnage, engaged in the carriage of liquefied gases having a vapour pressure exceeding 0.28 MPa absolute at a temperature of 37.8°C, and other products, as shown in chapter 19, when carried in bulk".
1.2 A ship carrying liquefied hydrogen in bulk (hereinafter called "liquefied hydrogen
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products, as shown in chapter 19, when carried in bulk".

1.2 A ship carrying liquefied hydrogen in bulk (hereinafter called "liquefied hydrogen carrier") should comply with the Code.
1.3 The Code requires that a gas carrier should comply with the minimum requirements for the cargo listed in chapter 19. However, the requirements for liquefied hydrogen are not specified in the Code.
1.4 This annex provides the Revised Interim Recommendations, as referred to in paragraph 5 of the preamble of the Code, for the carriage of liquefied hydrogen in bulk, which are intended to provide the basis for the future minimum requirements for the carriage of this cargo. The Revised Interim Recommendations are intended to facilitate the establishme nt of a tripartite agreement among the relevant Administrations for the carriage of liquefied hydrogen in bulk. However, they are not intended to prohibit the adoption of designs and arrangements
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in bulk. However, they are not intended to prohibit the adoption of designs and arrangements
other than those specified in the Code or in these Recommendations, at the discretion of the Administrations.
1.5 These recommendations have been developed under the assumption that a liquefied hydrogen carrier does not carry liquefied gases other than liquefied hydrogen. These recommendations, therefore, are not applicable to liquefied hydrogen carriers carrying gase s other than liquefied hydrogen.
1.6 In the Code, reference is made to paragraph 5 of the Preamble; paragraph 1.1.6.1; and Note No.8 on completion of certificate in "model form of international certificate of fitness for the carriage of liquefied gases in bulk" in appendix 2 to the Code.
1.7 These Revised Interim Recommendations consist of the following parts. Part A is applicable to ships with any type of cargo containment system. Part B and subsequent part(s)

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applicable to ships with any type of cargo containment system. Part B and subsequent part(s)
prescribe additional special requirements for cargo containment systems of specific types.
Part A: General (applicable to ships with any type of cargo containment system);
Part B: Cargo containment systems of independent cargo tanks using vacuum insulation; and
Part C: Cargo containment systems of independent cargo tanks using insulation materials and hydrogen gas in the inner insulation spaces.
RESOLUTION MSC.565(108) (adopted on 24 May 2024)
REVISED INTERIM RECOMMENDATIONS FOR CARRIAGE OF LIQUEFIED HYDROGEN IN BULK

1.8 Part A of this document was developed based on the design of parts B and C. If subsequent part(s) are added, the special requirements prescribed in part A may be reviewed.
Part A
General
(Applicable to ships with any type of cargo containment system)
2 GENERAL

2.1 Definition

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reviewed.
Part A
General
(Applicable to ships with any type of cargo containment system)
2 GENERAL
2.1 Definition
2.1.1 The following definition should apply for the purpose of these Revised Interim Recommendations.
Permeation is flow of a fluid through another material by diffusion without a defect or opening of the latter.8
2.2 Requirements for carriage of liquefied hydrogen in bulk
2.2.1 The requirements for the carriage of liquefied hydrogen in bulk have been developed based on the results of a comparison study of similar cargoes listed in chapter 19 of the Code, e.g. liquefied natural gas.
2.2.2 Chapter 19 of the Code governs the application of general requirements for respective cargoes. Selections of the general requirements for respective cargoes are expressed in columns 'c' to 'g'. In addition to general requirements, special requirement s may apply to specific cargoes depending on the properties/hazards of the cargoes.

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specific cargoes depending on the properties/hazards of the cargoes.
2.2.3 Tables 1 and 2 specify the proposed selection of the general requirements and the special requirements, respectively, for liquefied hydrogen. In addition to table 2, special requirements for cargo containment systems of specific types are prescribed in part B or subsequent part(s).
8 1 See paragraph 3.79 of ISO/TR 15916:2015.